

Supporting Information

Reversible β -Sheet-like Ordering and XPCS-Resolved Dynamic Arrest during Elastin-like Polypeptide Phase Separation

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48 1 Recombinant Synthesis and Purification of (VPAVG)₃₀

49 1.1 Materials

50 Shrimp alkaline phosphatase, PfuUltra II Fusion HS DNA polymerase, GoTaq Green Master Mix, and
 51 T4 DNA ligase with PEG4000 additive were purchased from Thermo Fisher Scientific. Exonuclease I
 52 was purchased from New England Biolabs. CircLigase ssDNA ligase was purchased from Lucigen, and
 53 KOD Hot Start DNA polymerase was purchased from Sigma-Aldrich. PCR products and plasmids were
 54 purified using Monarch PCR and DNA Cleanup kits and Monarch Plasmid Miniprep columns (New
 55 England Biolabs). DNA was recovered from agarose gels using the Monarch DNA Gel Extraction kit (New
 56 England Biolabs). Oligonucleotides and the 5'-phosphorylated synthetic template used for rolling-circle
 57 amplification were obtained from Integrated DNA Technologies. *E. coli* BL21(DE3)-Gold cells were
 58 obtained from Stratagene.

59 1.2 Gene construction

60 The ELP gene was prepared by overlap-extension rolling-circle amplification (OERCA).[1] A single-
 61 stranded DNA template encoding five VPAVG repeats was circularized in a 20 μ L reaction containing
 62 10 pmol template, 2 μ L of 10 $\times$ CircLigase buffer, 1 μ L of 1 mmol L⁻¹ ATP, 1 μ L of 50 mmol L⁻¹ MnCl₂,
 63 and 100 U CircLigase. The reaction was incubated for 1 h at 60 °C, heat-inactivated for 10 min at 80 °C,

purified, and treated with exonuclease I for 1 h at 37 °C to remove unreacted linear template.

OERCA was performed with 300 ng circular template, 1.5 μL each of 50 $\mu\text{mol L}^{-1}$ primers A1 and A2 (Figure S1a), 2 μL of 70% GC dNTP mix, 2 μL of 25 mmol L^{-1} MgSO_4 , 10 μL of 10 $\times$ PfuUltra II buffer, 5 μL DMSO, and 2 μL PfuUltra II polymerase in a total volume of 100 μL . Cycling consisted of 2 min at 95 °C; 30 cycles of 20 s at 95 °C, 20 s at 54 °C, and 90 s at 72 °C; and a final 10 min at 72 °C. Products above approximately 1.2 kb were isolated from a 0.8% agarose gel.

Ligation-independent cloning adapters were introduced by PCR using primers B1 and B2 (Figure S1a), and the products were inserted into the pMCSG53 vector.[2, 3] Constructs were transformed into *E. coli* BL21(DE3)-Gold, screened for soluble expression, and sequence verified by Sanger sequencing. The selected construct encodes the amino-acid sequence MHHHHHSSGVDLGTENLYFQSNA (VPAVG)₃₀, comprising a cleavable N-terminal affinity/fusion sequence followed by 30 VPAVG repeats. That leader was kept rather than cleaved, because its single tyrosine is the only chromophore in the chain and the A_{280} concentration measurement depends on it. Every measurement reported here is therefore on the tagged construct, and the leader accounts for 24 of the 174 residues and holds all eight charged residues.

1.3 Protein expression and purification

Fresh transformants were grown in Terrific Broth containing 10 mmol L^{-1} MgSO_4 and 300 $\mu\text{g mL}^{-1}$ ampicillin at 37 °C to an optical density of 0.6–0.8 at 600 nm. Cultures were cooled to 18 °C, induced with 0.2 mmol L^{-1} IPTG, and grown overnight. Cells were harvested at 5000 $\times g$, resuspended in ice-cold lysis buffer (500 mmol L^{-1} NaCl, 5 vol% glycerol, 50 mmol L^{-1} 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES) at pH 8.0, 20 mmol L^{-1} imidazole, and 10 mmol L^{-1} 2-mercaptoethanol), treated with 1 mg mL^{-1} lysozyme for 10 min, sonicated, and clarified at 29,000 $\times g$. [4]

The supernatant was applied to 5 mL IMAC Sepharose High Performance resin, washed with lysis buffer, and eluted with lysis buffer containing 500 mmol L^{-1} imidazole. Purification fractions were monitored by SDS-PAGE. The purified protein was exchanged into 20 mmol L^{-1} HEPES at pH 8.0, 250 mmol L^{-1} NaCl, and 2 mmol L^{-1} dithiothreitol (DTT) and concentrated using a 10 kDa molecular-weight-cutoff centrifugal concentrator. Protein concentration was determined by direct A_{280} measurement on a NanoDrop spectrophotometer using the sequence-specific molar extinction coefficient applied during sample preparation (calculated molecular weight 15,461 Da and $\epsilon_{280} = 1490 \text{ M}^{-1} \text{ cm}^{-1}$, both for the tagged construct). The final concentration was approximately 100 mg mL^{-1} ; samples were stored at 4 °C, and freshly prepared protein was used for the X-ray measurements. The expression, solubility, and purity of the protein were evaluated by SDS-PAGE (Figure S1b).

2 Sample Environments and X-ray Measurement Setup

2.1 Temperature-dependent SAXS/WAXS at 12-ID-B

Temperature-dependent SAXS/WAXS was performed at Advanced Photon Source (APS) beamline 12-ID-B. Two aliquots of (VPAVG)₃₀ at approximately 100 mg mL^{-1} in 20 mmol L^{-1} HEPES (pH 8), 250 mmol L^{-1} NaCl and 2 mmol L^{-1} DTT, nominally the same concentration used for the 8-ID-I measurements, were sealed in 2 mm quartz capillaries. The two beamtimes fell in different APS run cycles several months apart, and because the protein is purified fresh for each experiment, the 12-ID-B and 8-ID-I material came from separate preparations of the same construct. The aliquot designated Measurement

underwent more than 10 heating/cooling cycles across the thermal transition, whereas the Reference aliquot remained refrigerated at 4 °C. Both aliquots were subsequently measured at 10 °C and 30 °C during APS run 2021-1. The 13.3 keV beam had an approximate size of $100 \times 150 \mu\text{m}^2$ and an incident flux of approximately 5×10^{11} photons s^{-1} . SAXS was recorded with an Eiger2 9M detector at an approximately 2 m sample-to-detector distance, the same energy and camera length used for a published SAXS measurement at this beamline[5] and the same camera length as another[6], while WAXS was recorded with a Pilatus 300K detector at 0.42 m. The Pilatus detector was mounted above the direct-beam path, allowing the transmitted beam to continue to the SAXS detector so that SAXS and WAXS were acquired simultaneously with a 0.1 s exposure. Buffer and capillary backgrounds were subtracted, and the SAXS and WAXS profiles were stitched over their common Q interval. At each set point the capillary was held at the target temperature for at least 10 min before the exposures were started, so each profile reports the structure present after at least that long at temperature rather than during the temperature change. The two experiments used different sample cells, thermal paths, and thermometry, and the time spent at 30 °C before each 12-ID-B exposure is known only as a lower bound. The 12-ID-B 30 °C profiles are therefore kept off the isothermal waiting-time axis of the 8-ID-I experiment; they establish the low- and high-temperature structures and their reversibility, not the kinetics of assembly.

2.2 Isothermal SA-XPCS at 8-ID-I

Small-angle X-ray photon correlation spectroscopy (SA-XPCS) was performed at APS 8-ID-I. Approximately 100 mg mL^{-1} (VPAVG)₃₀ in the same buffer was loaded into an aluminum cell containing a 3 mm-diameter, 2 mm-thick liquid chamber sealed by 0.127 mm polycarbonate windows (Figure S2). The cell was inserted into an anodized-aluminum holder in one temperature zone of the Quantum Northwest 3×3 multi-sample temperature control unit, the same stage used in Ref. 7. Each zone is driven by a Peltier element and a resistive heater, and a calibrated resistance-temperature detector (RTD) embedded adjacent to the Peltier stage supplies the controller feedback and readout. The aluminum holder and aluminum cell provided direct conductive thermal coupling. A partially coherent 10.94 keV X-ray beam with 0.03% FWHM bandwidth and incident flux 1.2×10^{10} photons s^{-1} was focused to approximately $10 \times 10 \mu\text{m}^2$. Coherent scattering was recorded 7.8 m downstream with a Rigaku XSPA-500k photon-counting detector comprising 1024×512 pixels with a $76 \mu\text{m}$ pixel size and a $320 \mu\text{m}$ -thick silicon sensor. The detector was operated in 2-bit mode at a 50 kHz frame rate; each acquisition contained 100,000 frames, corresponding to 20 μs per frame and 2 s total duration. The multi-tau delay list is built from the recorded 20 μs frame time, and the longest accessible delay is 1.638 s.[8]

For the isothermal experiment, the loaded sample was first held at 6 °C for more than 30 min. It was then heated to 20 °C at $5 \text{ }^\circ\text{C min}^{-1}$, followed by a slower ramp from 20 °C to the target temperature of 30 °C at $1 \text{ }^\circ\text{C min}^{-1}$. The two-stage protocol was used to reduce overshoot near the target temperature. The waiting time t_w was set to zero when the calibrated RTD readout reached 30 °C; data collected during the preceding ramps were not included in the isothermal waiting-time series. From that point the measurement simply ran. Single 2 s acquisitions were taken one after another for the rest of the experiment, each at a previously unexposed position on a grid of sample positions, at a cadence of roughly 6.3 s per acquisition once stage motion and readout are included. A single acquisition is therefore a snapshot of a fresh piece of sample at a known t_w , and the series steps through increasing t_w at that cadence. SAXS profiles were obtained by temporal and azimuthal averaging. Each reported elapsed time is a group average, not a single acquisition: consecutive 2 s acquisitions were averaged in contiguous run ranges of 63–200 acquisitions after cross-correlation outlier rejection, and the quoted t_w is the acquisition time of the first run in the range. The five groups analyzed for g_2 begin at $t_w = 5040, 5970, 6596, 7222$ and

7863 s and hold 150, 100, 100, 100 and 63 acquisitions, spanning 15.5, 10.4, 10.4, 10.7 and 6.7 min; the last is short only because the measurement ended there. The $t_w = 0$ SAXS profile of Figure 3a averages the first 200 acquisitions, over 20.8 min. Within any one group the fitted fast fraction (Section 5.3) changes by less than its own group-to-group step, so this averaging does not materially broaden the correlation functions. Correlation functions were extracted at these elapsed times once the low- Q intensity provided sufficient photon statistics.

3 12-ID-B SAXS/WAXS Analysis

3.1 Background subtraction and SAXS/WAXS merging

The beamline-reduced .avg files contain Q , $I(Q)$ and a pointwise intensity uncertainty, the counting error $\sqrt{I}$. For each aliquot and temperature the capillary-plus-buffer profile was subtracted with a scale factor α of 1.00 for SAXS and the empirical 0.95 used in the original WAXS reduction, and the uncertainties were propagated through that subtraction as $\sigma = \sqrt{\sigma_s^2 + \alpha^2 \sigma_b^2}$. The WAXS intensity and its uncertainty were then multiplied by the ratio of the mean SAXS and WAXS intensities over their common Q interval, that ratio being treated as exact, and the two profiles were concatenated and sorted by Q .

Those propagated uncertainties enter one result. The Guinier fit of Section 3.3 is weighted, each point carrying $1/\sigma_{\ln I}^2$ with $\sigma_{\ln I} = \sigma_I/I$, and they therefore set the quoted errors on R_g and I_0 . The low- Q power-law fit and the two Gaussian peak fits are unweighted, so the exponent and the peak centers and widths do not depend on them. Neither the 0.95 nor the overlap ratio affects a reported result either: no absolute WAXS intensity is used anywhere in this work, and the centers and widths of the two peaks, and hence the β -sheet-like spacings, depend on neither.

3.2 Low- Q power law and WAXS peak fits

For Figure 2 of the main text, the low- Q slope was obtained by an unweighted linear fit of $\log_{10} I$ against $\log_{10} Q$ over $0.012 < Q < 0.040 \text{ \AA}^{-1}$ for the Reference aliquot at 30 °C. The two WAXS features were fitted independently to a Gaussian plus linear background over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70 \text{ \AA}^{-1}$, respectively. The low- Q exponent is reported as an exponent and is deliberately not converted into a fractal dimension. Two conditions set out by Teixeira[9] are not met. First, a mass-fractal dimension must satisfy $D < 3$, whereas the fitted 12-ID-B slopes are -3.17 and -3.15 ; exponents between 3 and 4 belong instead to the surface-fractal form $Q^{-(6-D_s)}$, which that reference notes is not easily distinguished from the Q^{-4} of a smooth interface. Second, a fractal description is expected to hold over at least one decade in Q , and the fit window $0.012 < Q < 0.040 \text{ \AA}^{-1}$ spans 0.52 decades. The isothermal 8-ID-I exponents, -2.45 to -2.68 , do fall below 3, so the two datasets sit on opposite sides of that boundary; this is one more reason they are treated as complementary rather than as replicate measurements of one state. The reduced profiles and scripts implementing the background subtraction, overlap scaling, Guinier analysis, power-law fit, and peak fits are available in the public repository identified in Section 8.

3.3 Low-temperature unassembled state

The Reference profile at 10 °C contains a Guinier regime satisfying $Q_{\max}R_g = 1.246$ (Figure S3). A fit of $\ln I(Q)$ against Q^2 , weighted by those propagated uncertainties, gives

$$R_g = 22.65 \pm 0.50 \text{ \AA}. \quad (\text{S1})$$

For reference, the cross- β pattern is reported at approximately 4.7–4.8 Å along the fibril direction and between 8 and 11 Å perpendicular to it.[10, 11] Our 8.8 Å falls inside that second range and our 4.5 Å sits just below the first.

The fitted size is molecular in scale and is consistent with a dispersed, unassembled low-temperature state rather than persistent mesoscale aggregates. Because the protein concentration is high, this value is reported as an apparent R_g ; weak intermolecular structure-factor contributions cannot be excluded. Recovery of the same low-temperature profile after repeated cycling establishes that the large high-temperature assembly is no longer detectable after cooling.

4 8-ID-I Isothermal SAXS

4.1 Absolute scattering cross section

For the 8-ID-I data, the differential scattering cross section was calculated as

$$I(Q) = A_s I_s(Q) - A_b I_b(Q), \quad (\text{S2})$$

where $I_s(Q)$ and $I_b(Q)$ are the temporal-mean detector intensities from the sample and buffer, per pixel and per frame. The conversion factors were

$$A_j = \frac{N_f}{F_j T_j t_{\text{exp}} L \Delta\Omega}, \quad j \in \{s, b\}, \quad (\text{S3})$$

where F_j is the incident photon flux, T_j is the transmission, $N_f = 100,000$ is the number of frames, $t_{\text{exp}} = 2 \text{ s}$ is the total exposure time, so that t_{exp}/N_f is the 20 μs frame time, $L = 2 \text{ mm}$ is the sample thickness, and $\Delta\Omega = (d_{\text{pix}}/D)^2$ is the solid angle per detector pixel.[12] Lengths are carried in millimeters throughout, so $A_j I_j$ is in mm^{-1} , and that is the quantity plotted in Figures 3a, S4b and S5: no further scaling of any kind is applied between the calibration and the figure.

4.2 Transmission and incident-flux calibration

Sample transmission was obtained from ion chambers immediately upstream and downstream of the sample. Across the attenuation series, the transmission of the intervening air path was independent of beam intensity within uncertainty, with a mean value of 0.8679 ± 0.0009 (Figure S6a; mean $\pm$ standard deviation). The upstream ion-chamber signal was converted to incident photon flux by calibration against a PIN diode positioned in the flight path upstream of the beamstop. The measured photon flux varied linearly with the upstream ion-chamber readout (Figure S6b). The absolute-scale SAXS evolution in Figure S5 used this transmission and flux calibration.

4.3 Isothermal SAXS evolution at 30 °C

Figure S5 shows the full SAXS sequence. At $t_w = 0$, the profile is close to the 6 °C reference. The low- Q intensity then increases continuously and ultimately by more than two orders of magnitude, documenting progressive growth of mesoscale density contrast. The profile remains broad throughout, consistent with hierarchical network development rather than selection of a narrow domain spacing. The most pronounced increase occurs between $t_w = 4089$ and 5040 s, after which the intensity continues to grow through 7863 s. This growth is what eventually brings the low- Q coherent intensity above the level at which a two-step correlation function can be resolved, and is therefore the origin of the $t_w \geq 5040$ s restriction. The correlation analysis in Figure 3 of the main text uses $t_w = 5040$ – 7863 s; earlier scattering was insufficient for reliable two-step correlations at the $20 \mu\text{s}$ frame time. The interval common to both beamlines is $0.012 < Q < 0.033 \text{ \AA}^{-1}$, narrower at the top end than the $0.012 < Q < 0.040 \text{ \AA}^{-1}$ range behind the slopes quoted in the main text. Over that common interval the 12-ID-B 30 °C profiles give power-law slopes of -3.09 (Reference) and -3.07 (Measurement), whereas the isothermal 8-ID-I series steepens from -1.07 at $t_w = 0$ through -1.78 at 4089 s to -2.45 at 5040 s, and then stops. The last four groups give -2.675 , -2.670 , -2.642 and -2.644 , a spread of 0.03 across the final 1894 s, against a step of 0.23 in the interval immediately before. Over those same 32 min the intensity averaged over $0.004 < Q < 0.008 \text{ \AA}^{-1}$ still grows by a factor of 1.7 , so the low- Q scattering keeps gaining amplitude at an essentially fixed shape. Within the window measured, then, the isothermal series is not simply en route to the 12-ID-B exponent: its shape stopped evolving about half an hour before the series ended, roughly 0.4 short of -3.08 . The two datasets are therefore treated as complementary rather than as replicate measurements of one state, and the gap between them is not ascribed to waiting time alone, the cells, thermal paths and preparations all being different.

5 8-ID-I SA-XPCS Correlation Analysis

5.1 Correlation functions and the fitted model

The intensity autocorrelation function was calculated with a multi-tau algorithm,

$$g_2(Q, \Delta t) = \frac{\langle I(Q, t)I(Q, t + \Delta t) \rangle_t}{\langle I(Q, t) \rangle_t^2}. \quad (\text{S4})$$

The implementation is the 8-ID-I reduction workflow; its numerical treatment of the multi-tau scheme at these frame rates is described in detail elsewhere.[8, 13]

The reduction produced 27 azimuthally averaged dynamic Q bins of nominal width $1.13 \times 10^{-3} \text{ \AA}^{-1}$ spanning 0.00376 – $0.03307 \text{ \AA}^{-1}$. The five lowest bins, centered at $Q = 0.00376$, 0.00489 , 0.00602 , 0.00714 and $0.00827 \text{ \AA}^{-1}$ and corresponding to $2\pi/Q$ of approximately 167 , 128 , 104 , 88 and 76 nm, are analyzed here. Both this Q restriction and the restriction to $t_w \geq 5040$ s are signal-to-noise limits: the scattered intensity falls by more than an order of magnitude between the lowest bin and $Q = 0.00827 \text{ \AA}^{-1}$ (Figure S5), so above that bin the correlation function is dominated by counting noise, and before $t_w = 5040$ s the low- Q intensity is itself too weak to resolve the second decay step. Figure S4c shows the still-mobile limit of that behavior directly.

The measured g_2 carry two decays separated by several decades in delay, and were parameterized as a

sum of two stretched exponentials,

$$g_2(Q, \Delta t) = 1 + \beta \left[f(Q) \exp \left[- \left(\frac{\Delta t}{\tau_{\text{fast}}(Q)} \right)^{p_{\text{fast}}} \right] + [1 - f(Q)] \exp \left[- \left(\frac{\Delta t}{\tau_{\text{slow}}(Q)} \right)^{p_{\text{slow}}} \right] \right]^2. \quad (\text{S5})$$

This is the same functional form used for thermoreversible nanoparticle gels, where the two-step decay is fitted as a linear sum of two stretched exponential modes with complementary amplitudes a and $1 - a$.^[14] The same two-mode decomposition, with a fast decay and a slow stretched decay whose amplitude sets the intermediate plateau, describes ergodicity breaking in aging nanoclay suspensions.^[15] Here $f(Q)$ is the fitted fraction of fast dynamics and $1 - f(Q)$ its complement. $\tau_{\text{slow}}(Q)$ is free at each Q ; τ_{fast} is not, being tied to the diffusive form $\tau_{\text{fast}}(Q) = 1/(D_0 Q^2)$ so that one coefficient replaces five free times. The Kohlrausch–Williams–Watts exponents carry no Q argument because they are shared across the five bins, and p_{fast} is shared across the elapsed times as well. The series of five is therefore described by 61 parameters rather than the 125 a fully free fit would need. The decomposition is phenomenological: it describes the observed first decay and persistent plateau without assigning either mode to a unique molecular mechanism.

5.2 Calibration of the speckle contrast from a static reference

The contrast β of Eq. S5 is set by the transverse coherence of the incident beam and by the ratio of speckle size to detector pixel size. It is a property of the beamline configuration rather than of the sample state, so it is measured once and then held fixed. The measurement used a static reference, a nano-porous glass standard of 10 nm nominal pore diameter recorded in the same configuration shortly before the isothermal series.

A static sample does not decorrelate, so $g_2(\Delta t) = 1 + \beta$. The reference was reduced on the same terms as every other group, 41 of its 50 acquisitions surviving the outlier rejection described in Section 2.2. For each of the 27 Q bins those 41 were averaged, and a constant was fitted to that average in $\log \Delta t$ by weighted least squares using the pointwise g_2 uncertainties; the level of the constant is $1 + \beta$. The adopted value is the mean of that constant over the five Q bins the sample itself is analyzed in, so that β is measured in the range it is applied in. The five are flat, with reduced χ^2 between 0.71 and 1.03 and levels of 0.1295, 0.1312, 0.1329, 0.1315 and 0.1319 (Figure S7a shows the lowest of them). Their mean is

$$\beta = 0.13139 \pm 0.00056. \quad (\text{S6})$$

The quoted error is the standard error of that mean. It is used rather than the error of any single constant fit, which comes out near 10^{-5} : the multi-tau delay points share frames and are strongly correlated, so a fit that treats them as independent understates its own uncertainty, and the scatter between bins is the realistic measure of how well β is known.

Repeating the procedure bin by bin gives $\beta(Q)$ (Figure S7b). Twenty-three of the 27 bins return a flat averaged correlation function and a usable level; four do not. β is flat at approximately 0.132 out to $Q \approx 0.021 \text{ \AA}^{-1}$ and falls beyond it, reaching 0.107 in the outermost bin. That falloff is instrumental: the 0.03% bandwidth transmitted by the Si(111) monochromator sets a finite longitudinal coherence length, and the path-length difference across the scattering volume grows with scattering angle, so contrast is progressively lost as Q increases. The five bins analyzed for the sample all lie below that onset. The strongest-counted flat bin anywhere, $Q = 0.02067 \text{ \AA}^{-1}$, returns 0.13031, within one standard error of Equation S6, and the bin-to-bin scatter falls as $1/\sqrt{N}$ with the number of acquisitions averaged, so it is counting noise rather than a Q dependence. Equation S6 is the value used throughout, with the baseline fixed at 1.

5.3 Global fitting and uncertainties

Equation S5 was fitted to all five Q bins of one elapsed time at once, with p_{fast} and p_{slow} shared between the bins and f , τ_{fast} and τ_{slow} free at each Q . Fits minimized the error-weighted residuals $[g_{2,\text{model}} - g_{2,\text{data}}]/\sigma_{g_2}$ with `scipy.optimize.least_squares`, using the pointwise g_2 uncertainties stored in the reduced files. Bounds were wide ($0 \leq f \leq 1$, $10^{-6} \leq \tau_{\text{fast}} \leq 10$ s, $1 \leq \tau_{\text{slow}} \leq 10^4$ s, $0.2 \leq p \leq 3$) and no parameter reaches one. The five global fits give reduced $\chi^2 = 0.78\text{--}1.89$. Starting values and the remaining numerical settings are in the code (Section 8).

Parameter uncertainties are the square roots of the diagonal of the covariance matrix, estimated as the Moore–Penrose pseudo-inverse of $J^T J$ with J the Jacobian of the error-weighted residuals at the solution; the pseudo-inverse is used so that a parameter refining to a bound does not leave the uncertainties of the remaining, well-constrained parameters undefined. These are the error bars plotted in Figure 3c. The significance of the Q dependence of f follows from them, the difference between two bins being compared with $\sigma = \sqrt{\sigma_i^2 + \sigma_j^2}$: for adjacent bins at $t_w = 5040$ s the four differences are 2.0 , 2.2 , 2.9 and 1.8σ , and at the four later times the smallest adjacent difference is 8.6 , 11.5 , 13.4 and 7.3σ . Taken end to end across the five bins the difference is 9.0σ or more at every elapsed time. Power-law relations $\tau = A Q^\gamma$ in Figure S8 were obtained by weighted linear regression of $\ln \tau$ against $\ln Q$, with $\sigma_{\ln \tau} = \sigma_\tau / \tau$.

The slow component remains only partially relaxed during the 2 s acquisition and frequently yields $\tau_{\text{slow}} > 2$ s, so it is an effective extrapolation rather than a sampled terminal relaxation time. Conclusions about arrest therefore rest on directly observed features: the decreasing amplitude of the first decay, the rising plateau, and the systematic decrease in f . XPCS does not determine where the arrested regions are, $1 - f$ is not a geometric volume fraction because scattering contributions are weighted by structure and contrast, and because the measurement does not reach infinite delay it is not the formal nonergodicity factor either. Figure S8 reports all auxiliary fit parameters, and Figure S9 the correlation functions and fits at the four additional Q values: the same two-step shape and the same monotonic loss of the fast step appear at every measured Q , so the trend of Figure 3c belongs to the whole 76–167 nm range and not to a single bin.

5.4 Length-scale-dependent relaxation parameters

Figure S8 shows the slow-mode parameters from the joint fit. Both stretching exponents remain below unity: $p_{\text{fast}} = 0.595 \pm 0.023$, fitted once for the whole waiting-time series, and $p_{\text{slow}} = 0.21\text{--}0.48$, free at each elapsed time. Fitting p_{fast} at each elapsed time instead returns 0.42 to 0.68 and changes reduced χ^2 from 1.195 to 1.188, an improvement on four extra parameters significant only at $p = 0.015$ by an F test; the χ^2 profile is flat to within 1% between 0.55 and 0.70, so a single value is what the data support. These stretched decays indicate a broad distribution of relaxation times rather than a single process. Heterogeneous, nonequilibrium dynamics of this kind are characteristic of protein and colloidal gels,[14, 16] although the direction of the departure from $p = 1$ varies between systems: the egg-white gel of Ref. 16 relaxes with compressed rather than stretched exponentials, the signature of stress-driven rather than diffusive relaxation. Stretching of this kind is consistent with diffusion: a Kohlrausch g_2 still describes a diffusive system provided p is independent of Q , which holds here by construction, and τ scales as Q^{-2} , which an unconstrained fit confirms below.

The fast relaxation occurs on approximately $1 \times 10^{-4}\text{--}2 \times 10^{-3}$ s and becomes faster with increasing Q as Q^{-2} , which Equation 2 imposes rather than fits. Releasing that constraint, so that τ_{fast} is free in every bin, returns Q exponents of -2.08 to -2.37 at the four elapsed times where the fast mode is

resolved, which is why the diffusive form is the one reported. The fitted coefficients are $D_{\text{eff}} = 3.36 \pm 0.41$, 4.69 ± 0.29 , 5.28 ± 0.36 , 3.34 ± 0.24 and 1.82 ± 0.35 ($\times 10^7 \text{ \AA}^2 \text{ s}^{-1}$) in waiting-time order. At the final elapsed time the fast amplitude at the lowest Q bin has fallen to $f = 0.002$ and its relaxation time is no longer measured there. The fitted slow component also decreases with Q , but its tens-to-thousands-of-seconds values exceed the acquisition window; neither τ_{slow} nor γ_{slow} is interpreted as a directly measured terminal relaxation law. The robust arrest measure remains the monotonic decrease in the fitted fast fraction f across all five Q values, equivalently the increase in the fitted slow fraction $1 - f$.

6 Radiation-Damage Mitigation and Controls

6.1 Fresh sample positions

The grid of positions referred to in Section 2.2 was Cartesian, 100×100 points in the aluminum cell, with adjacent points separated by at least one beam diameter. Because no position was visited twice, cumulative dose cannot track the elapsed time of the isothermal experiment, so the trend in the fitted fast fraction cannot be a dose artifact.

6.2 Flux-dependence control

In addition, to show that there is no significant radiation-induced artifact, an independent dose-rate test was performed after the APS Upgrade using a 20 mg mL^{-1} (VPAVG)₃₀ sample. Every acquisition was taken at a stage-RTD readout of 33.0°C , above the transition temperature, so the control probed the assembled high-temperature state rather than the low-temperature dispersed one. Attenuation selected incident fluxes from 2.7×10^{11} to $4.9 \times 10^{11} \text{ photons s}^{-1}$, approximately 23–41 times the flux used for the main experiment. The SAXS line shapes and g_2 decays showed no systematic flux dependence (Figure S10). That control sample is five times more dilute than the 100 mg mL^{-1} material of the main experiment, and it was measured with the post-Upgrade beam. The main series is protected instead by the fresh position used for every acquisition (Section 6.1).

7 Macroscopic and Mesoscale Structural Reversibility

7.1 Macroscopic reversibility

Video S1 provides a direct visual demonstration of thermal reversibility. Brief warming of a capillary containing (VPAVG)₃₀ produces turbidity within approximately 5 s, indicating structures large enough to scatter visible light. After removal of the heat source, the sample returns to optical transparency in approximately 60 s. This recovery is consistent with the overlap of the cycled Measurement and refrigerated Reference SAXS/WAXS profiles at 10°C . Optical transparency does not specify molecular structure, but independently confirms dissolution of the mesoscale assembly on cooling.

7.2 The seven-cycle experiment

The reversibility comparison of Figure 2 sets two separate 12-ID-B aliquots against each other. The test in Figure S4 is more direct: one aliquot at 8-ID-I, taken through seven consecutive cycles and compared

with itself, so that any difference between cycles can only be irreversibility. The aliquot is not the loading used for the isothermal experiment, but comes from the same stock and was measured two days apart in the same cell position of the same holder, with the cell reloaded in between. Each cycle held the sample at 6 °C, ramped to 34 °C at 1.01 °C min⁻¹ over 270 consecutive 2 s acquisitions, and returned to 6 °C, taking 5.8 h in total. No acquisitions were taken during cooling, so the cooling segments of Figure S4a are a projection from the fixed control protocol rather than a measurement; every solid segment is measured, once per acquisition, from the RTD of the stage zone holding the cell.

7.3 Cycle-to-cycle repeatability

Three states are sampled in every cycle: 6 °C, and two windows near the top of each ramp at 31.9 °C and 33.8 °C. Averaged over $0.004 < Q < 0.008 \text{ \AA}^{-1}$, the seven cycles agree to a relative standard deviation of 9% at 6 °C and 5% at 33.8 °C, with no drift from the first cycle to the seventh. At 31.9 °C the spread is 33%, but that reflects where on the ramp each cycle happened to be sampled rather than a difference between cycles: the windows fall between 31.62 °C and 32.48 °C, which is on the steep part of the transition. Regressing the band-averaged $\ln I$ of each cycle on that cycle's mean window temperature gives $d \ln I / dT = 0.95 \text{ }^{\circ}\text{C}^{-1}$ with $r = 0.974$, and the spread taken about that line falls to 7%. The same regression at 33.8 °C, past the steep part, takes 5% to 1%. The cycles therefore repeat to better than 10% at every state once the sampling temperature is accounted for.

7.4 Dynamics at the cycled states

The correlation functions say what state the sample was in when each was taken (Figure S4c). At 33.8 °C, g_2 stays close to $1 + \beta$ across the whole accessible delay range in every one of the seven cycles: the sample gelled on every heating. It does not quite reach $1 + \beta$, and the shortfall is a measurement rather than an instrumental defect. Averaged over the seven cycles the level is 0.1209, 0.1192, 0.1156, 0.1118 and 0.1065 across the five Q bins, against the bin-by-bin instrumental values of Figure S7b, 0.1295, 0.1312, 0.1329, 0.1315 and 0.1319. Read through Equation S5 with the slow mode taken as fully arrested, that shortfall is a residual fast fraction of 0.03, 0.05, 0.07, 0.08 and 0.10, against 0.01, 0.04, 0.08, 0.10 and 0.13 for the isothermal series at $t_w = 7863 \text{ s}$ (Figure 3c). A separate loading, brought to 33.8 °C on a ramp rather than held isothermally, therefore retains motion of the same size and with the same increase toward high Q as the isothermal series does at the end of its hold. At 31.9 °C the correlation decays essentially to the baseline in the same window, so the assembly is still mobile 1.9 °C lower. That is the direct evidence behind the elapsed-time restriction used in the main text: while the sample is still mobile the low- Q correlation completes a single decay inside the window, leaving no second step to resolve. No correlation function is shown at 6 °C, where the coherent signal at the lowest Q is too weak for any contrast to be resolved.

7.5 Cold-state artifacts and the outlier cut

Two features of the 6 °C state need comment. First, its profile still rises toward low Q instead of reaching the flat Guinier plateau of a fully dispersed solution. Mis-subtracted solvent does not account for it: the buffer contributes only about 35% of the signal at the lowest Q , so rescaling it to remove the rise would subtract most of the low- Q signal and would degrade the cycle-to-cycle agreement rather than improve it. Second, the cold state is where transient scatterers appear. Taking every acquisition as recorded, the intensity at $Q = 0.0035 \text{ \AA}^{-1}$ within a single 6 °C group is strongly right-skewed, while the same

acquisitions at $Q = 0.03 \text{ \AA}^{-1}$ are symmetric at ordinary counting statistics; a long positive tail confined to low Q is the signature of a large object crossing the $10 \times 10 \text{ \mu m}^2$ beam during a minority of acquisitions. The cross-correlation cut does not catch them, because it compares curve shape over the whole Q range and a curve anomalous in three low- Q bins alone stays close to the group mean by that measure.

A second, robust cut removes them, and is applied to every group in the paper on identical terms with no per-group list. For each acquisition the mean of $I(Q)$ over $0.004 < Q < 0.008 \text{ \AA}^{-1}$ is compared with the group median, and the acquisition is rejected if it exceeds the median by more than three robust standard deviations, the scale being $1.4826 \times$ the median absolute deviation and both statistics iterated to convergence. The test is one-sided because a transient scatterer only adds intensity. It removes 35 of the 1749 acquisitions surviving the cross-correlation cut, 2.0%: 4, 2, 1, 11, 4, 1 and 1 from the seven 6°C groups, 2 and 7 from the two buffer groups, and none at all from any of the fourteen high-temperature windows, which is consistent with the skew being confined to the cold state. Elsewhere it is nearly inert, removing two acquisitions of the isothermal series, shifting one absolute-scale coefficient by 0.03% and leaving every fitted correlation-function parameter unchanged. Its effect here is reported rather than absorbed: the cycle-to-cycle spread at 6°C falls from 13% to 9%, and what remains is a mild rise over the last two cycles rather than one anomalous group. The threshold is not tuned, the same acquisitions being selected for any value between 2.5 and 3 and for two other choices of band.

One difference between the two loadings survives all of this and is a property of the material rather than of a few acquisitions: the cycling loading has $I(0.004)/I(0.02) = 4.0$ against 2.7 for the isothermal loading, and transmits 0.305 against 0.333. It was the less completely dissolved of the two.

8 Code, Data and AI-Assisted Development

8.1 Code and reduced-data availability

The reduced datasets and all analysis and figure-generation code used for this manuscript are publicly available at <https://github.com/qzhang234/PubCode-2021-IDPP>. The repository includes the reduced 12-ID-B SAXS/WAXS data, the averaged 8-ID-I XPCS HDF files for every group plotted, the per-acquisition temperature record of the thermal-cycling sequence, radiation-control data, absolute-intensity calibration inputs, and the scripts used to generate the main-text and Supporting Information figures. Every data-derived figure can be regenerated from the repository alone. Two further scripts document how the committed reduced files were produced from the raw beamline results. `SAXS_12id/Read_12ID_SAWAXS.py` handles 12-ID-B, merging the SAXS and WAXS profiles into the committed CSV files. `SAXPCS_8id/average_ranges.py` handles 8-ID-I, applying both outlier-rejection passes to every group on the same terms and averaging each into the committed HDF files. Both read the beamline storage directly. The repository README maps each script and output file to the corresponding manuscript figure. The raw detector frames occupy 3.0 TB and the reprocessed per-acquisition results 51 GB, both held on Advanced Photon Source project storage inside the facility network. The 69 MB of averaged files committed here are what the figures are produced from. Raw data are available from the corresponding authors.

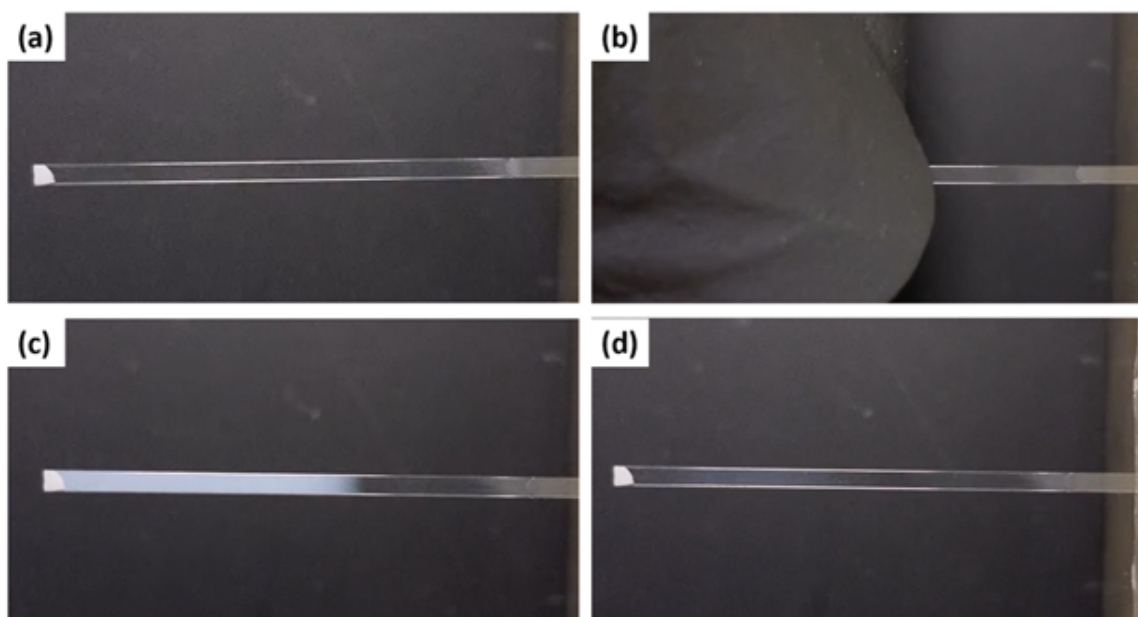
8.2 AI-assisted development

The authors used Claude Sonnet 5 and Opus 5 to assist with the manuscript preparation and code development. The Claude models were accessed through Argo, Argonne National Laboratory's internal

generative AI interface, which provides Laboratory staff with access to large language models from multiple vendors, including Anthropic, OpenAI, and Google, for use in Laboratory research and operations.

Qingteng Zhang reviewed the code, reran the analyses, and confirmed the reported results. The code is public (Section 8.1), so every reported number and data-derived figure can be checked independently. All analysis choices, scientific interpretations, and manuscript content were determined by the authors, who take full responsibility for the publication.

Supporting Video



Video S1. Reversible thermal transition in (VPAVG)₃₀. (a) A thin quartz capillary containing (VPAVG)₃₀ is held in a fixture. (b) The capillary is warmed for approximately 5 s by contact through nitrile gloves. (c) The initially transparent solution becomes visibly turbid. (d) The sample returns to transparency approximately 60 s after removal of the heat source. The complete video can be viewed or downloaded at <https://anl.app.box.com/s/lsukldjnr45nhfrkobij sar1tqhba7jw>.

(a)

```

A1 (forward)
5'-GTTCCAGCTGTTGGTG

A2 (reverse)
5'-ACCTACTGCTGGTACG

B1 (forward)
5'-TACTTCCAATCCAATGCCGTTCCAGCTGTTGGTGTGCCG

B2 (reverse)
5'-TTATCCACTTCCAATGTTAACCTACTGCTGGTACGCCGAC

VPAVG30 OERCA single-stranded template
/5Phos/GTTCCAGCTGTTGGTGTGCCGGCCGTGGGCGTGCCTGC
GGTGGGCGTCCCGGCGGTCCGCGTACCAGCAGTAGGT

```

(b)

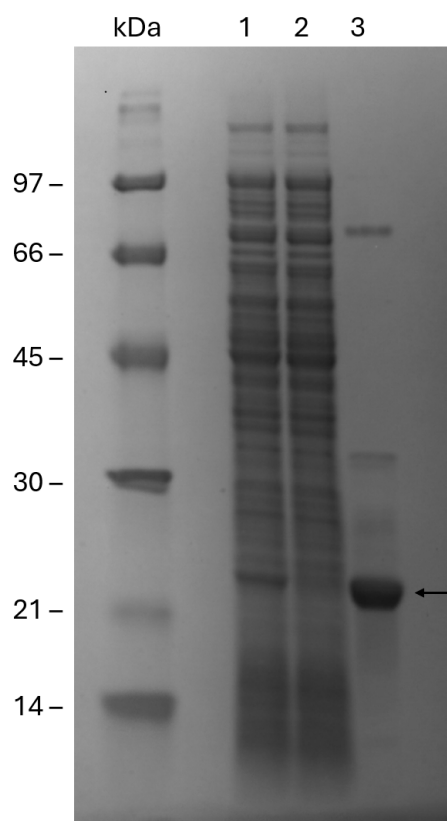


Figure S1: Construction and expression of the (VPAVG)₃₀ gene. (a) Oligonucleotide primers and the 5'-phosphorylated single-stranded OERCA template used to construct the ELP gene. (b) SDS-PAGE analysis of expression and purification. Samples were resolved by SDS-PAGE and stained to evaluate expression, solubility, and purity of the heterologously expressed protein. The molecular-weight marker lane is at the left, with prominent bands labeled in kilodaltons (kDa). Lanes 1–3 are the total soluble fraction, the flow-through fraction after IMAC, and the eluted fraction, respectively. The arrow marks the eluted (VPAVG)₃₀ construct, whose apparent mobility is lower than that expected from its calculated mass of 15,461 Da.

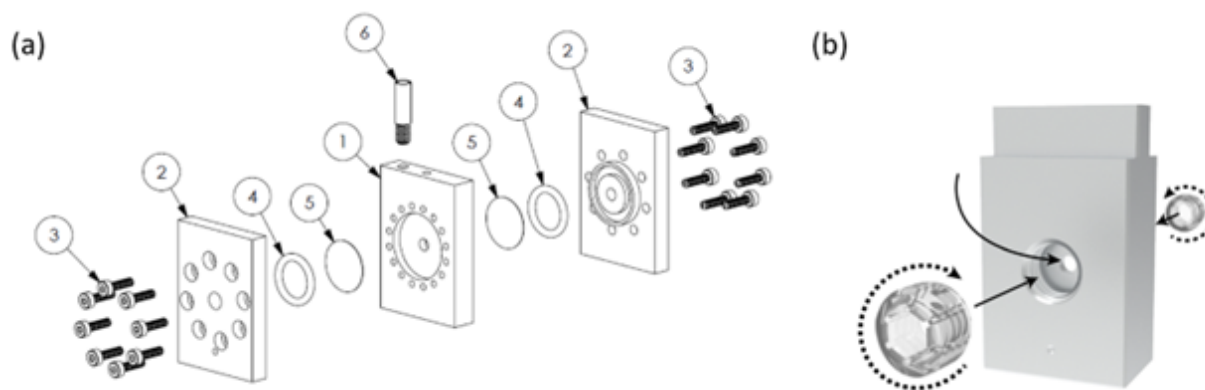


Figure S2: Sample cells used for liquid X-ray measurements. (a) Exploded view of the aluminum cell used for the 8-ID-I isothermal SA-XPCS experiment. The cylindrical chamber is 3 mm in diameter and 2 mm thick along the X-ray direction; 0.127 mm polycarbonate windows are compressed by Viton O-rings and bolted caps. During measurement, this cell was inserted into the anodized-aluminum holder of the Peltier-controlled stage described in Section 2.2; both the holder and cell were aluminum. (b) Screw-cap cell developed separately for automated handling; the dashed circles magnify the window seat. Only the bolted aluminum cell in panel (a) was used for measurements reported in the main text. The circled numerals in (a) identify the components: (1) the aluminum cell body, which contains the sample chamber; (2) the aluminum cap plates; (3) M2 screws, 8 mm long, which clamp the caps against the body; (4) Viton O-rings; (5) window disks cut from Kapton or polycarbonate sheet with a hammer-driven punch, polycarbonate being used for the measurements reported here; and (6) the handle used to withdraw the cell from the Quantum Northwest stage. Labels (2)–(5) appear on both sides of the body because the assembly is symmetric about the sample chamber.

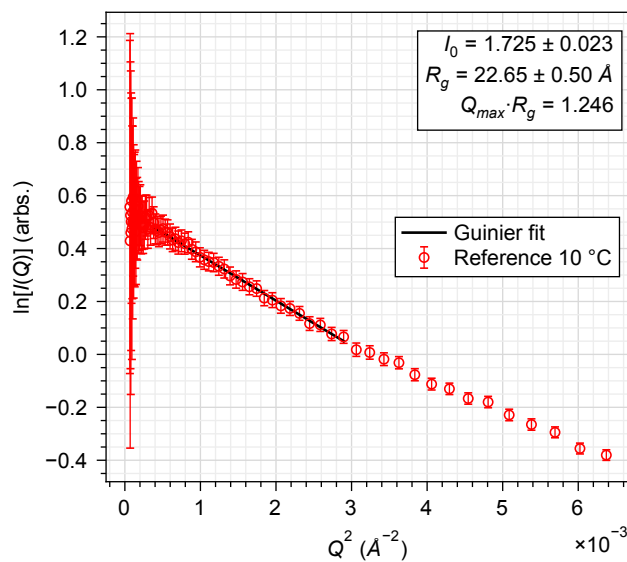


Figure S3: Guinier analysis of the Reference aliquot at 10 °C. Weighted linear fit of $\ln I(Q)$ against Q^2 over $0.020 \leq Q \leq 0.055 \text{ \AA}^{-1}$ (so $Q_{\max} R_g = 1.246$), giving an apparent $R_g = 22.65 \pm 0.50 \text{ \AA}$ and $I_0 = 1.725 \pm 0.023$ in the arbitrary intensity units of the reduced 12-ID-B profiles. Points are plotted over the wider range $0.005 \leq Q \leq 0.080 \text{ \AA}^{-1}$ for context; error bars are the propagated intensity uncertainties, σ_I/I .

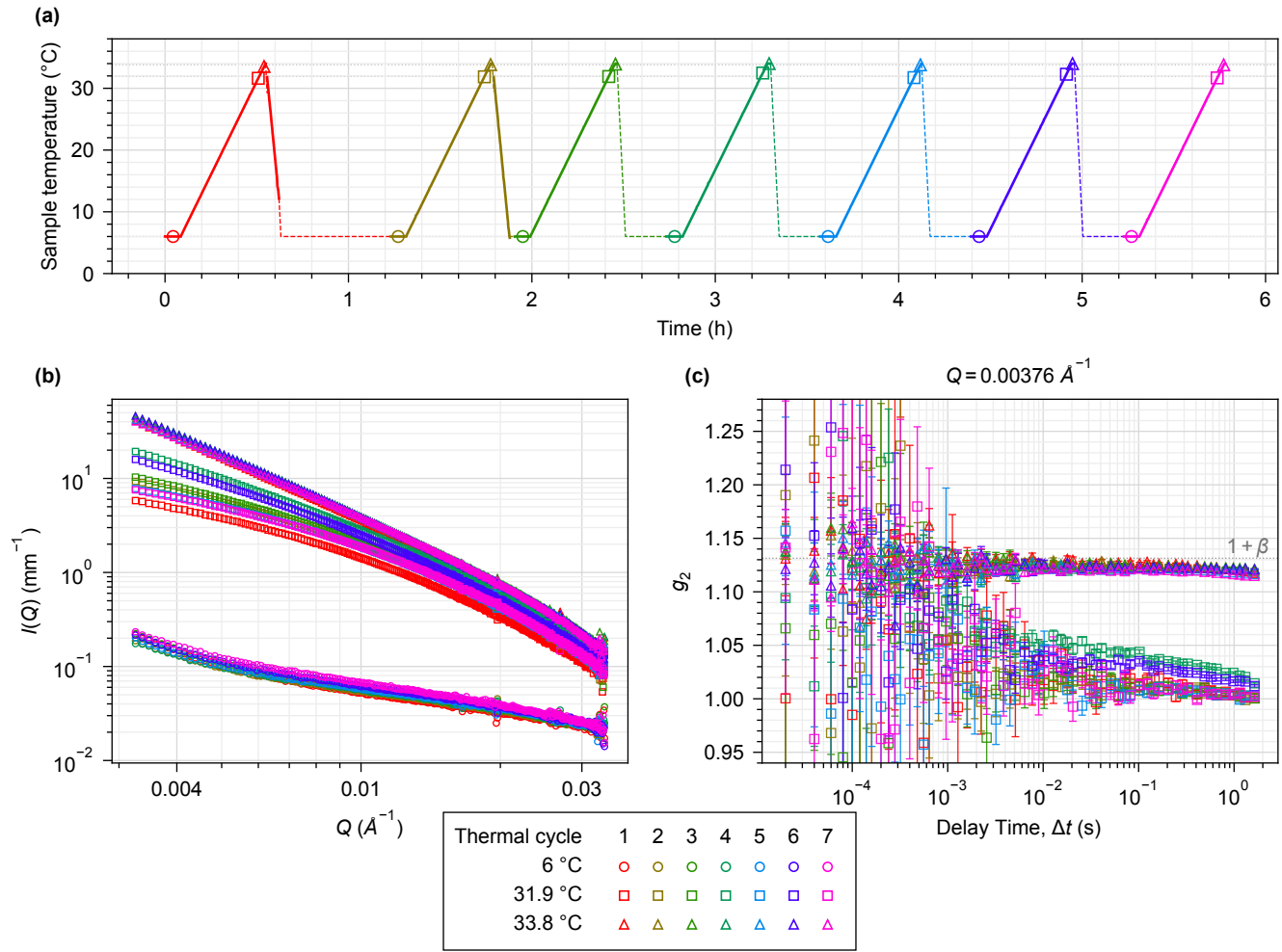


Figure S4: Thermal-cycling repeatability of the SA-XPCS sample: one aliquot, a separate loading from the isothermal experiment of Figure 3 of the main text, through seven consecutive cycles. (a) Sample temperature over the 5.8 h sequence. Solid curves are measured: the stage RTD was read once per 2 s acquisition. No acquisitions were taken while the sample cooled, so no temperature was recorded over those intervals; the dashed curves there are not data but the profile projected from the control protocol, which was fixed at 10 °C min⁻¹ cooling until 6 °C followed by a hold at 6 °C until the next ramp. Open symbols mark, at the time and temperature at which it was taken, every acquisition group that is plotted in (b) and (c), so each lower-panel curve can be traced to its point in the thermal history. (b) Buffer-subtracted SAXS on an absolute scale at 6 °C (circles, 32–41 surviving acquisitions averaged; see Section 7.2), 31.9 °C (squares, acquisitions 241–250) and 33.8 °C (triangles, acquisitions 261–270). Each group is placed on the absolute scale with the mean ion-chamber readings of the acquisitions it contains, and the averaged buffer (69 surviving of 89 acquisitions, D0077 and D0080) is subtracted with no empirical scale factor, exactly as in Figure 3a. (c) Correlation functions at the lowest Q bin for the two high-temperature states; the dotted line is $1 + \beta$ with β the contrast measured in Figure S7. Error bars are the measured pointwise g_2 uncertainties; the y axis is capped so that a few short-delay caps clip. One key serves all three panels: its columns are the seven thermal cycles, numbered along the top and colored on the same scale Figure 3 of the main text uses for elapsed time, and its rows are the three temperature states, labeled at the left and distinguished by marker shape.

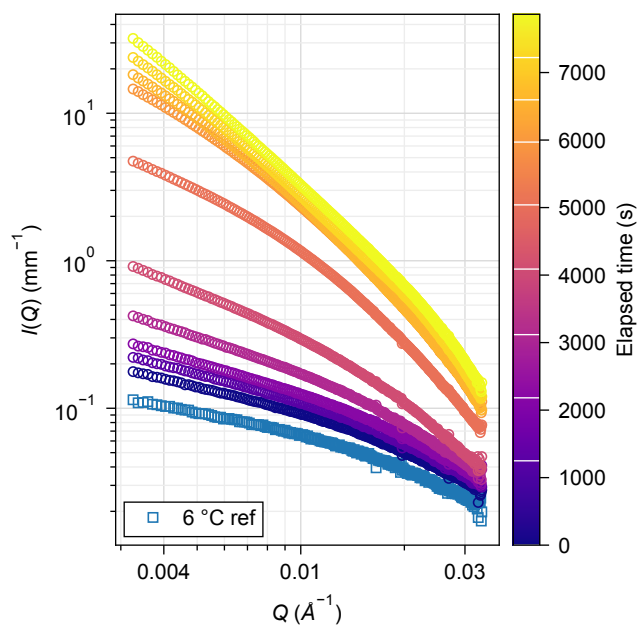


Figure S5: Complete isothermal SAXS evolution of (VPAVG)₃₀ at 30 °C, buffer-subtracted and on the same absolute scale as Figure 3a of the main text, using the same buffer measurement. The 6 °C reference and profile at the start of the hold are followed by profiles from 1249 to 7863 s. White marks on the color bar give the elapsed time of each measured profile. Colors here are scaled linearly over the full 0–7863 s range and therefore do not match the discrete colors used for the same times in Figure 3; read times from the color bar. The subset from 5040 to 7863 s corresponds to the SAXS and SA-XPCS data analyzed in Figure 3 of the main text.

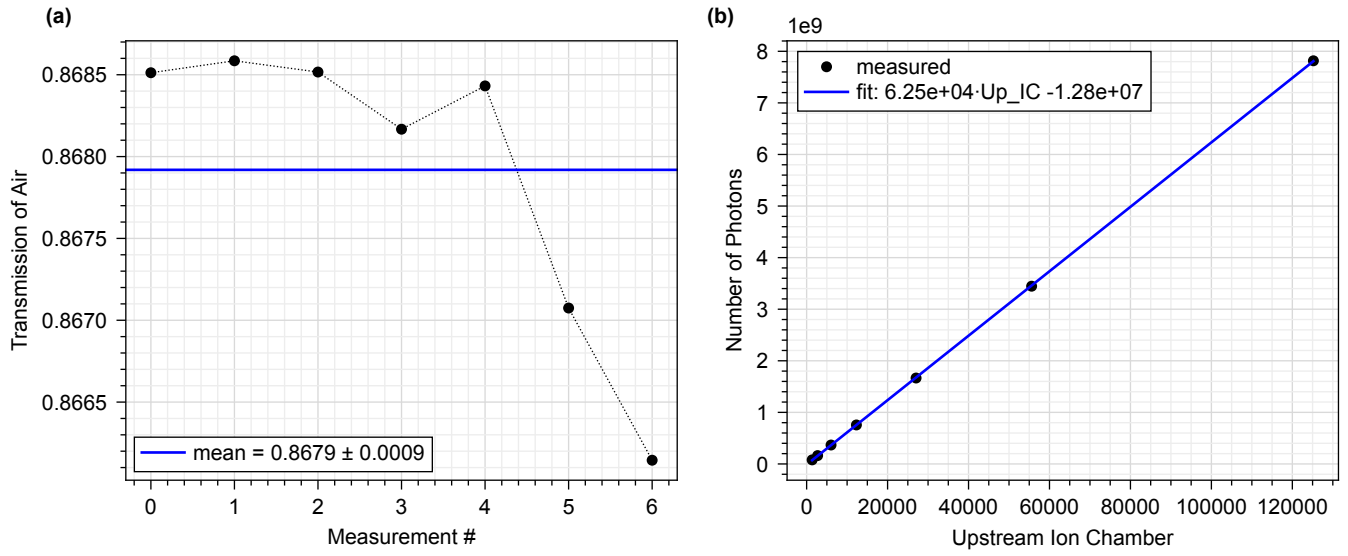


Figure S6: Transmission and photon-flux calibration for conversion to absolute scattering cross section. (a) Air-path transmission across seven measurements in the attenuation series; the line marks the mean, 0.8679 ± 0.0009 (standard deviation). (b) Photon counts recorded by a calibrated PIN diode as a function of the upstream ion-chamber readout (both in the scaled detector units of the calibration series); the line is the linear calibration fit, from which the incident flux follows as $F = (A \cdot \text{Up_IC})/t_{\text{exp}} + B$.

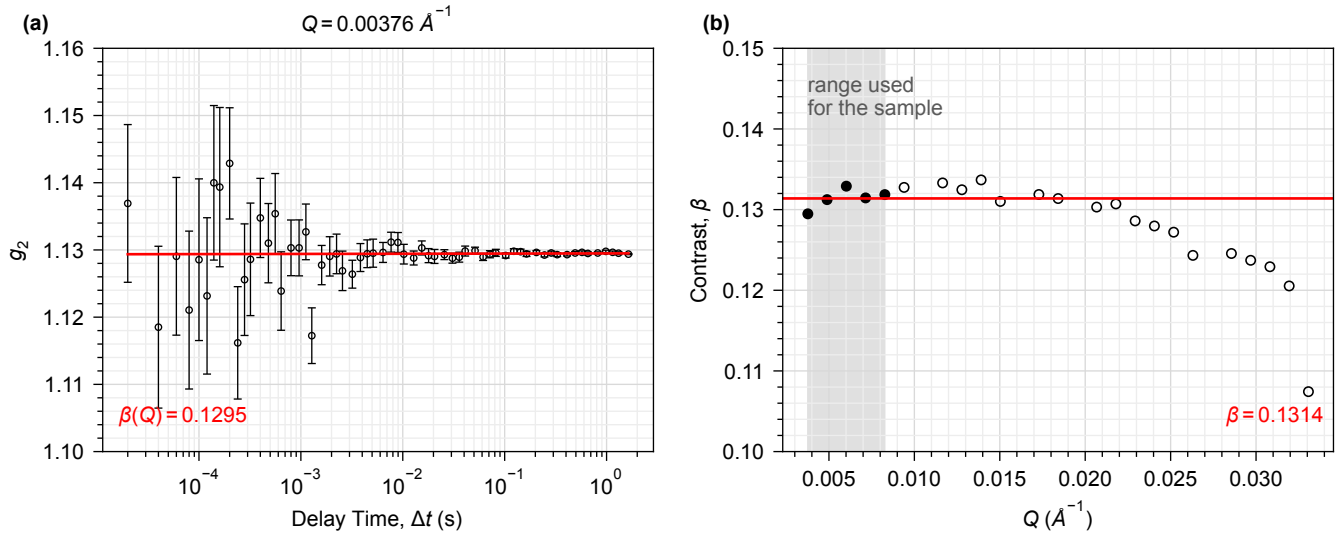


Figure S7: Measurement of the instrumental speckle contrast β on a static reference. (a) Averaged correlation function of a 10 nm nano-porous glass standard at $Q = 0.00376 \text{ \AA}^{-1}$, the lowest of the five bins the sample is analyzed in; error bars are the measured pointwise uncertainties. The red line is a straight-line fit in $\log \Delta t$; it is flat to 0.3 standard deviations with reduced $\chi^2 = 0.71$, as a static sample requires, and its level is 0.1295. (b) The same procedure applied bin by bin. β is essentially independent of Q across the shaded range used for the sample. Filled symbols are the five bins the sample is analyzed in and the red line is their mean, $\beta = 0.13139$, the value used throughout. Above $Q \approx 0.021 \text{ \AA}^{-1}$, outside that range, β falls off as the finite longitudinal coherence length set by the Si(111) monochromator begins to limit the contrast. The five return 0.1295, 0.1312, 0.1329, 0.1315 and 0.1319, and the strongest-counted flat bin anywhere, at $Q = 0.02067 \text{ \AA}^{-1}$, returns 0.1303, within one standard error of their mean. Four bins out of 27 whose averaged correlation function is not flat (reduced $\chi^2 > 5$) are omitted, and all four lie above the shaded range.

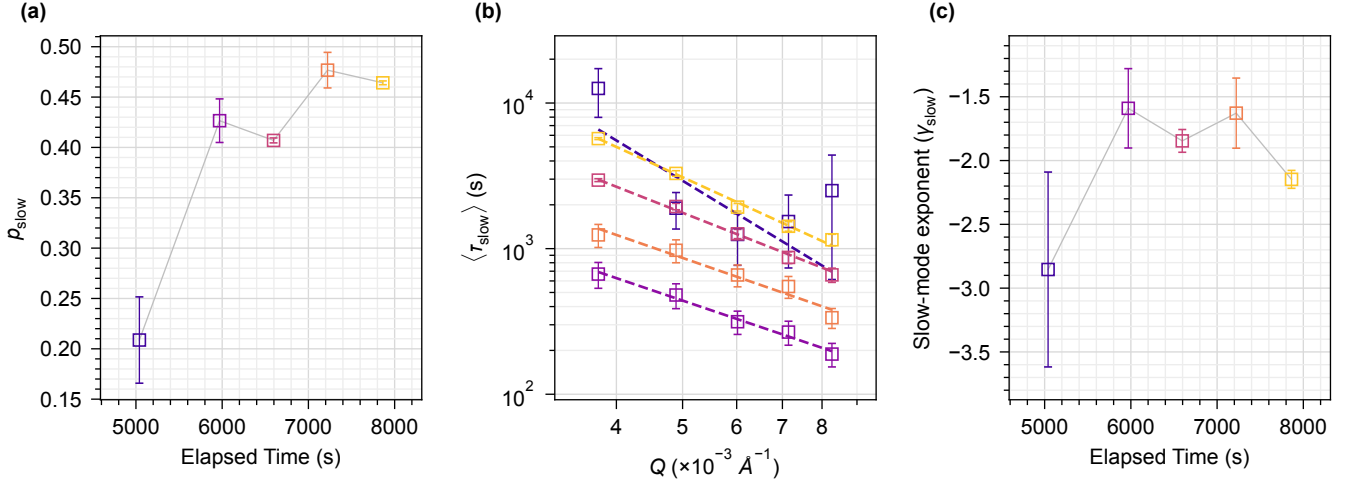


Figure S8: Slow-mode parameters from the joint fit of Equation 2. Colors denote elapsed time as in Figure 3 of the main text (5040, 5970, 6596, 7222 and 7863 s, dark to light). (a) The slow-mode stretching exponent p_{slow} , free at each elapsed time. (b) Mean slow relaxation time, $\langle \tau_{\text{slow}} \rangle = \Gamma(1/p_{\text{slow}}) \tau_{\text{slow}}/p_{\text{slow}}$, versus Q ; dashed lines are power-law fits $\langle \tau \rangle \propto Q^\gamma$. (c) The fitted exponents γ_{slow} . The fast mode has no panel here: Equation 2 ties it to $\tau_{\text{fast}}(Q) = 1/(D_0 Q^2)$, so plotting it against Q would reproduce that line and nothing more, and the one number it carries, D_{eff} , is in Figure 3c. $p_{\text{fast}} = 0.595 \pm 0.023$ is fitted once for the whole series. τ_{slow} exceeds the 2 s acquisition window, so these are extrapolated descriptors and are not used for mechanistic scaling.

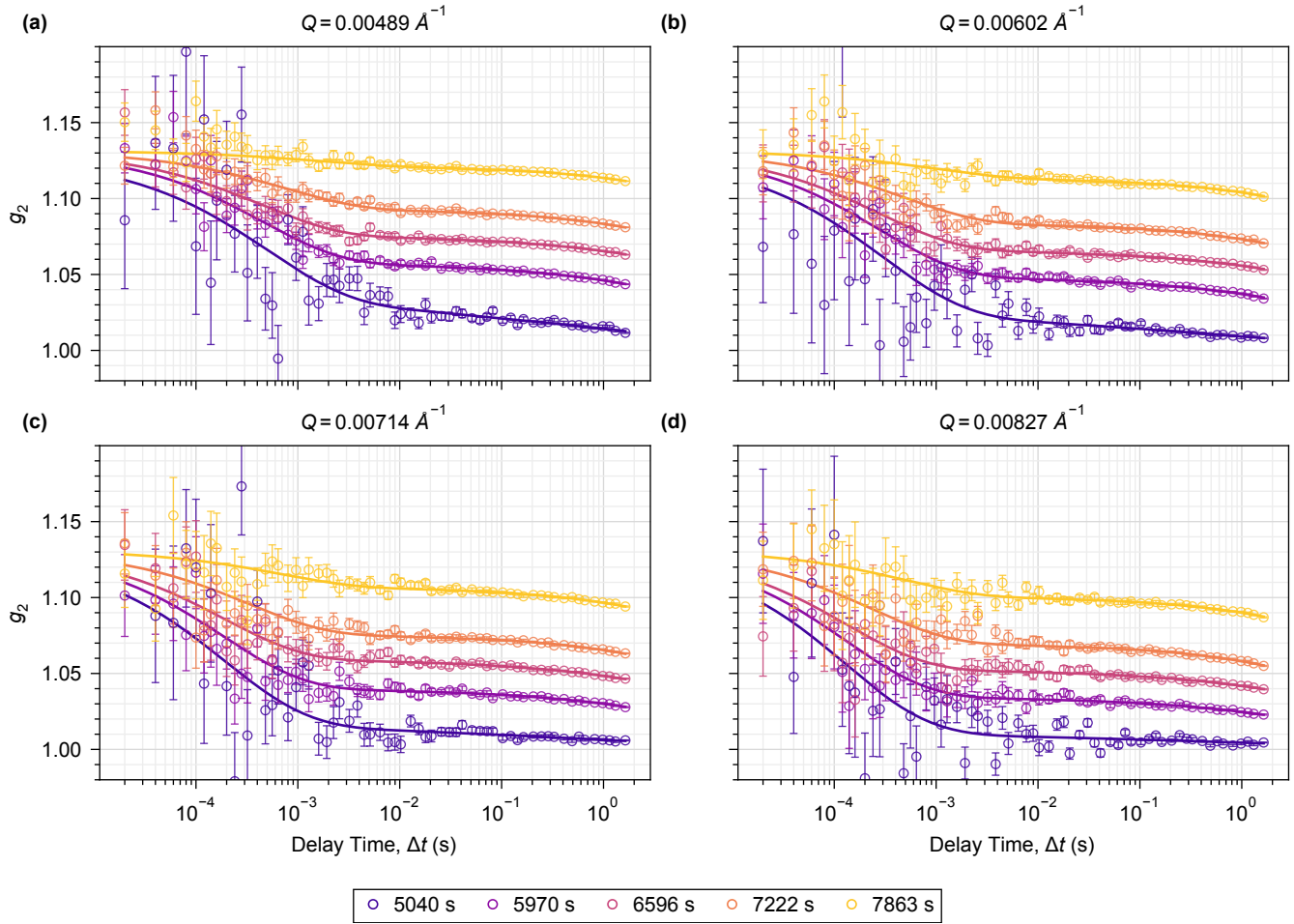


Figure S9: Expansion of Figure 3b of the main text onto the four higher Q bins. Panels show (a) $Q = 0.00489$, (b) 0.00602 , (c) 0.00714 , and (d) $0.00827 \text{ \AA}^{-1}$. Colors denote the same elapsed times as Figure 3b,c of the main text. Solid curves are the same global two-mode fits that produce Figure 3c and Figure S8: at each elapsed time all five Q bins are fitted simultaneously with p_{fast} and p_{slow} shared across Q and β fixed at the measured instrumental contrast, and the curve drawn in each panel is that global solution evaluated at that panel's Q . Error bars are the measured pointwise g_2 uncertainties. The y axis is capped at 1.20; two of 1220 plotted points and four error-bar caps, all at the shortest and noisiest delays, fall above it.

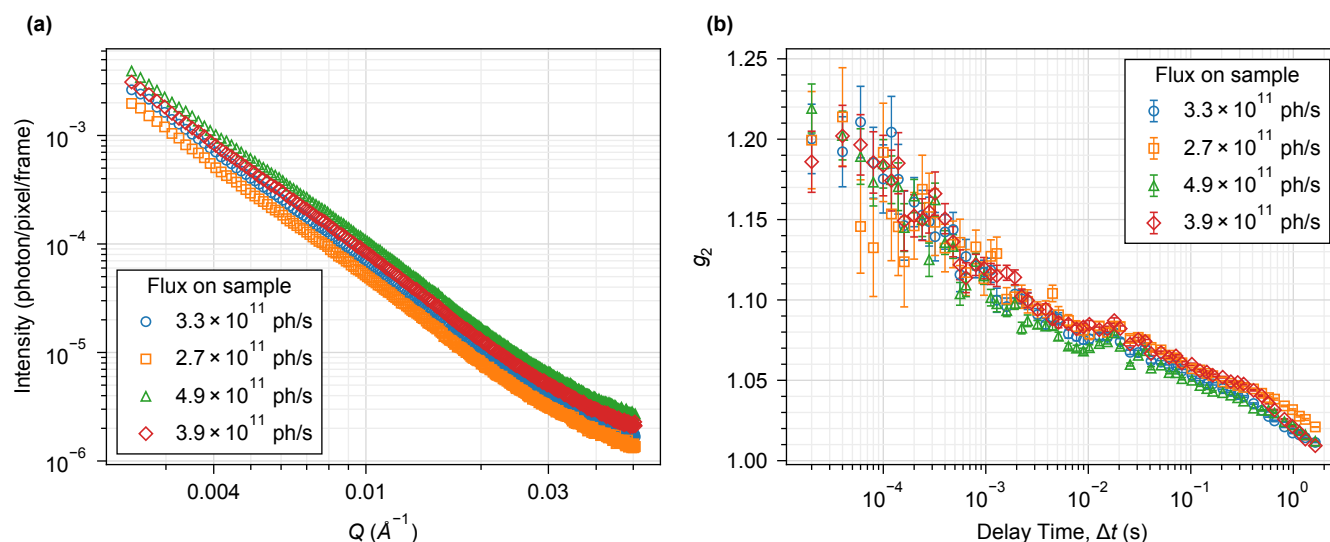


Figure S10: Flux-dependence control for SAXS and SA-XPCS. (a) SAXS profiles and (b) intensity autocorrelation functions from a 20 mg mL^{-1} (VPAVG)₃₀ sample at incident fluxes of 2.7×10^{11} – 4.9×10^{11} photons s^{-1} . The Q range shown in (a), 0.0024 – 0.051 Å^{-1} , covers the Q range of Figure 3a of the main text. Intensities in (a) are detector counts per pixel per frame, not on the absolute scale of Figure 3a, so the vertical spacing of the four curves is the incident flux itself: their mean intensities stand in the ratios 1.00, 1.29, 1.58 and 2.01 against incident-monitor ratios of 1.00, 1.24, 1.47 and 1.82. The line shapes nonetheless stay parallel across that range. Panel (b) shows the lowest Q bin of this control configuration, $Q = 0.00304 \text{ Å}^{-1}$ ($2\pi/Q \approx 206 \text{ nm}$). The legend orders the four conditions by nominal attenuator setting, which is not monotonic in flux. No systematic flux-dependent change is resolved over this range, which substantially exceeds the flux used in the main experiment.

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